The empirical recurrence for OEIS A186560, proved by transfer matrix

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Abstract

OEIS A186560 counts the $(n+2) \times 3$ arrays over $\{0, \dots, 2\}$ in which neighbouring 3×3 subblocks commute as matrices, and carries an empirical recurrence of order 7 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition ties together $3+1$ consecutive rows, so 3 consecutive rows are a state, the count is a walk count on $S = 19683$ of them, and the conjectured recurrence is then settled by a finite exact computation.

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1 The conjecture

OEIS A186560 is "Number of (n+2)X3 0..2 arrays with every 3X3 subblock commuting with each horizontal and vertical neighbor 3X3 subblock.". It has offset 1 and begins

$$19683, 721, 2917, 6594, 12122, 20056, 29303, 35198, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=2*a(n-1)-a(n-2)+2*a(n-3)-4*a(n-4)+2*a(n-5)+2*a(n-6)-2*a(n-7) for n>18

As of the "Last modified" line on the live entry (Oct 03 2025 03:35:40, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The contiguous 3×3 subblocks of an $(n+2) \times 3$ array form a grid of n rows and 1 column. Two subblocks that are neighbours in that grid OVERLAP: horizontally they share 3×2 entries, vertically 2×3. Nevertheless each is a 3×3 integer matrix in its own right, and the entry requires that every neighbouring pair, horizontal or vertical, commute — that is, $AB = BA$ as matrices over $\mathbf{Z}$.

At $n = 1$ the array has 3 rows and the grid is a single row of 1 subblock, with no neighbouring pair at all; the entry's first term is 19683, which the model reproduces along with every later term.

3 The count is a walk count

A subblock row is fixed by 3 consecutive array rows, and the vertical condition compares two consecutive subblock rows, hence $3 + 1$ consecutive array rows. So take as state the 3-tuple of consecutive rows whose subblock row has just been completed. A step appends one array row: that fixes the next subblock row, and the horizontal pairs inside it and the vertical pairs against the previous one are all settled there. An array of $n + 2$ rows is a walk of $n - 1$ steps and every walk comes from exactly one array, so

$$a(n) \;=\; \iota^\top M^{n-1}\tau, \qquad \iota = \tau = (1, \dots, 1)^\top,$$

on the $S = 19683$ admissible 3-tuples. In particular a is C-finite.

This state does not compress. What a step needs of the past is the previous subblock row, and for $3 \geq 3$ that row already determines the 3 array rows it was built from, so nothing is lost by carrying the rows and nothing would be gained by carrying the matrices instead. The size of the construction is therefore $3^{3 \cdot 3}$ before the condition prunes it, which is what puts the wider members of this family out of reach rather than settling them.

4 The proof

Lemma 1. *Let $q(t) = t^7 - \sum_i c_i t^{7-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+7} - \sum_i c_i M^{j+7-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 2\,a(n-1) - a(n-2) + 2\,a(n-3) - 4\,a(n-4) + 2\,a(n-5) + 2\,a(n-6) - 2\,a(n-7)$$

for every $n > 18$.

Proof. The vector $w = q(M)\tau$ was formed by 7 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 18 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 28 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A186560.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.